GALGOTIAS COLLEGE OF ENGINEEING & TECHNOLOGY (GCET)



Master of Computer Applications

Mini Project Review 1

Real Time Waste Reporting and Tracking System

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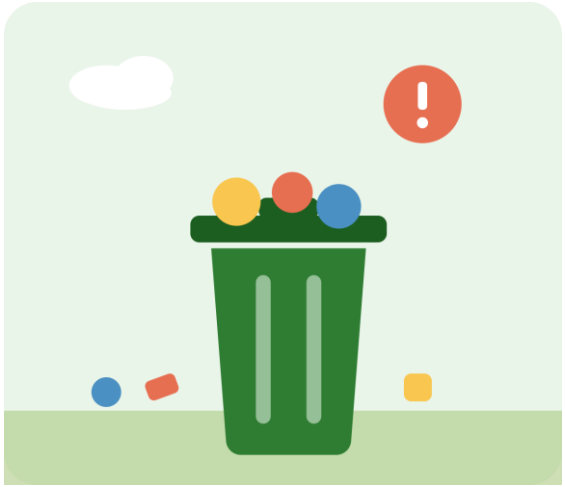
Real Time Waste Reporting and Tracking System

A simple web system to report waste and track its status



Introduction

- Waste collection in many areas is slow and not well organized.
- Complaints about garbage are often given by phone or on paper and are easy to lose.
- This project is a Django web application where users report waste and follow the status of their complaint.



Problem statement: Waste collection is not optimized, which increases time, fuel and operational cost.

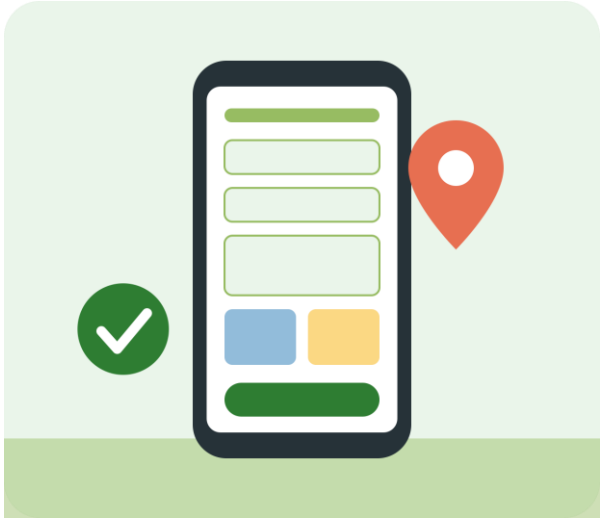
Problem statement

- Waste problems are often not reported properly.
- Manual complaint handling is slow and difficult to track.
- Citizens cannot easily check complaint status.
- Authorities face difficulty in managing multiple complaints.
- There is no centralized waste monitoring system.
- Our system helps users report, track, and manage waste complaints efficiently.

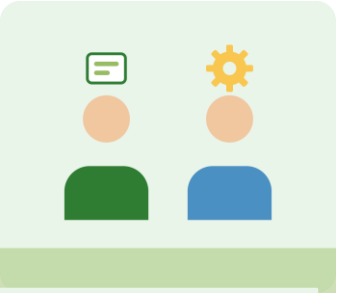


Objective

- Make it easy for people to report waste problems online.
- Collect waste type, location, description and an photo in one place.
- Give every complaint a unique ID and let users track its status.
- Help the admin view all complaints and update their status.
- Reduce delay, and save time, fuel and cost by keeping collection organized.



Scope



User

- Register and login
- Submit a waste complaint
- Choose waste type, add location and photo
- Get a complaint ID (e.g. WM-0001)
- View own complaints and status

Admin

- Login through Django admin
- View all complaints
- Search and filter by status or type
- Change complaint status
- Manage users and complaints

Complaint status: Pending | In Progress | Collected | Rejected

User Workflow

1. Register/Login – User creates an account or logs in.
2. Submit Complaint – User reports waste with type, location, description, and photo.
3. Get Complaint ID – System generates a unique complaint ID.
4. Track Status – User checks whether the complaint is pending, in progress, or collected.
5. Admin Management – Admin views, searches, filters, and updates complaints.
6. Waste Collection – Waste collection team resolves the complaint and collects the waste.



Key Features

- Online Waste Reporting – Users can report waste problems online.
- Unique Complaint ID – Every complaint gets a unique ID.
- Location & Photo Upload – Users can add location, waste type, description, and photo.
- Status Tracking – Track complaints as Pending, In Progress, Collected, or Rejected.
- Admin Management – Admin can view, search, filter, and update complaints.
- Efficient Waste Collection – Organized complaints help reduce delay, time, and fuel costs.



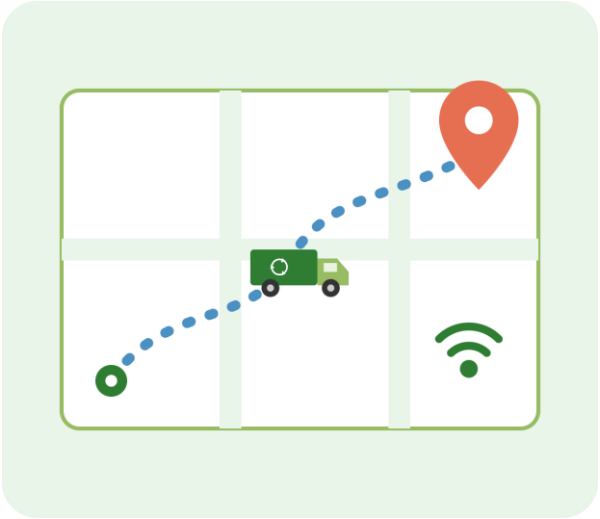
Benefits

- Easy and quick way to report waste from anywhere.
- Users can see whether a complaint is pending or collected.
- All complaints are stored in one place, so nothing is lost.
- Admin can find and handle complaints faster.
- Better planning helps save time, fuel and operational cost.
- Encourages responsible disposal and a cleaner environment.



Future Scope

- Add map and GPS location for each complaint.
- Add collector login and live tracking of collection vehicles.
- Suggest the best collection routes to save time and fuel.
- Send email or SMS updates when the status changes.
- Build a mobile app and an admin analytics dashboard.



Conclusion

- The system makes waste reporting simple and trackable.
- It keeps complaints organized and helps the admin act faster.
- With GPS, route planning and notifications it can become a full real-time waste tracking system.



Thank You